NAME:_____SCHOOL:_____

2020 **M131/I**

NSANJE DISTRICT MOCK

MALAWI SCHOOL CERTIFICATE OF EDUCATION EXAMINATION

MATHEMATICS

Subject Number: M131/I

Wednesday, 25 March

Time Allowed: 2 hours

8:00-10:00 am

PAPER I

(100 marks)

Instructions

1. **This paper contains 12 printed pages. Please check.**
2. Answer **all** the **20** questions in this paper.
3. The maximum number of marks for each answer is indicated against each question.
4. Write your answers in the spaces provided on the question paper.
5. Calculators may be used.
6. **All working must be clearly shown.**
7. Write your **Name and School Name** at the top of each page of your question paper in the spaces provided.
8. In the table provided on this page, **tick** against the question number you have answered.

Question Number	Tick if answered	Do not write in these columns	
1			
2			
3			
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20			
	Total		

1. Simplify $54_6 - 35_6$, giving your answer in base 4. (4 marks)

2. Given that $g(x) = x^2 - x - 1$, find $g(x + 1)$ (4 marks)

Continued/…

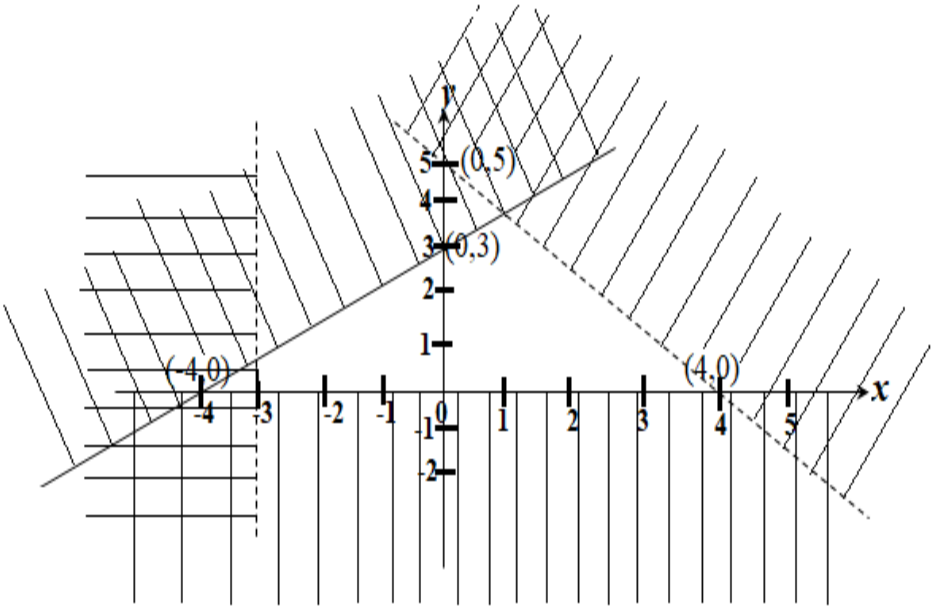
3. A regular polygon has interior angles of 150° each. Calculate the number of sides of the polygon. (4 marks)

4. Given that $B = N(1 + \frac{R}{X})^{K}$, make R the subject of the formula (5 marks)

Continued/…

5. Solve for y in $12 + 2^y = 2^{2+y}$. (4 marks)

6. **Figure 1** shows the region bounded by **four** inequalities.



Write down the four inequalities. (6 marks)

7. Anufa had mixed 80 kg of rice which was selling at K270 per kg with another type of rice which was selling at K210 per kg. If he had to sell a mixture at K250 per kg, how much of the second type of rice was used in the mixture? (5 marks)

8. Given that $25x^2 - 80x + V$ is a perfect square, find the value of V (5 marks)

Continued/…

9. Find the standard deviation for a set of data whose deviations from the mean are -2, -3, 4, 1 leaving your answer to 3 significant figures. (5 marks)

10. An arithmetic sequence with 16th term of 65 has a common difference of 4. Calculate the 20th term. (5 marks)

Continued/…

11. Given that $\begin{bmatrix} 2 & 8 \\ 4 & 4 \end{bmatrix}\begin{bmatrix} m & k \\ m & 2 \end{bmatrix} = \begin{bmatrix} 10 & 14 \\ 10 & 8 \end{bmatrix}$, find the values of **k** and **m**. (5 marks)

12. Construct a triangle **ABC** with angle **ABC**= 30°, **AB**= 4cm and **BC** = 6 cm. Draw a circumscribed circle of the triangle **ABC** and measure and state its radius. (6 marks)

Continued/…

13. **Figure 2** illustrates the motion of a body. Use it to answer the questions that follow.

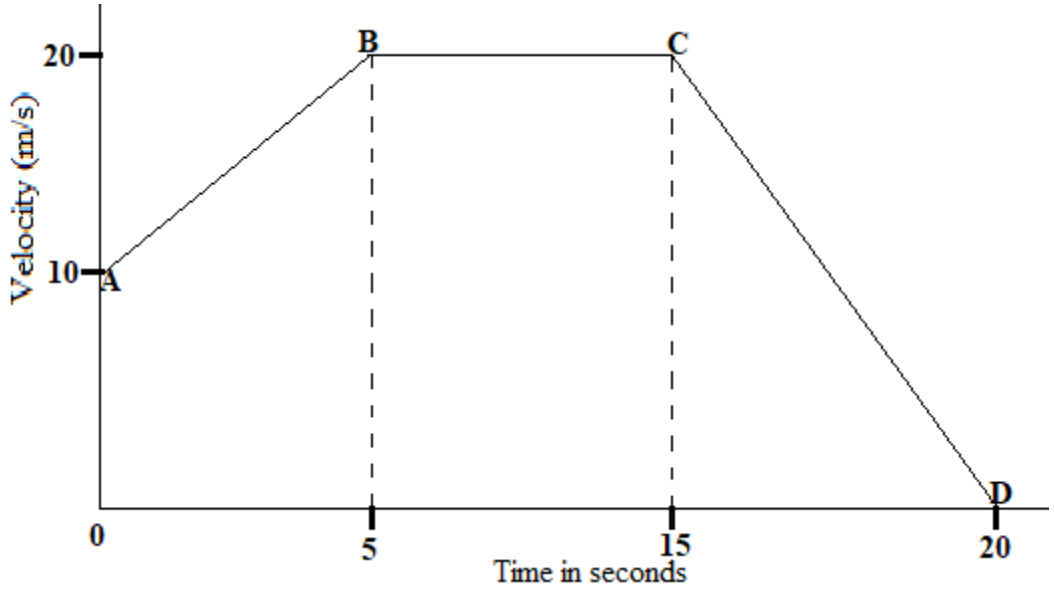


Figure 2

a. Calculate the acceleration from **A** to **B**. (3 marks)

b. Calculate the total distance travelled by the body (7 marks)

Continued/…

14. **MNOP** is a parallelogram. **H** is a point on **MN.** If **HO**= **NO**, prove that **MHOP** is a cyclic quadrilateral. (5 marks)

15. **Figure 3** shows a Venn diagram.

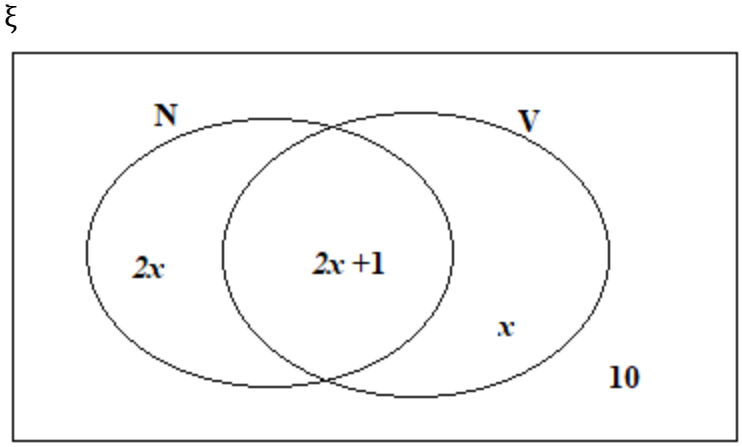


Figure 3

In the Venn diagram, ξ = {girls in form three}, N = {girls that play netball} V = {girls that play volleyball}. Given that there are 21 girls in the class, find how many girls play both netball and volleyball.

16. A certain translation maps **P** (3, 5) onto **P'**(7, 8). Find the coordinates of the point which is mapped onto **Q'**(2, 7) under same translation. (5 marks)

17. If $P \propto \frac{x^2}{z}$, P= 12 when z= 3, and *x*= 2. Find *x* when P= 30 and z= 5. (5 marks)

Continued/…

18. **Figure 4** shows a circle **PQRS** centre **O**. Angle **QOS** = 138° and **QSR**= 63°.

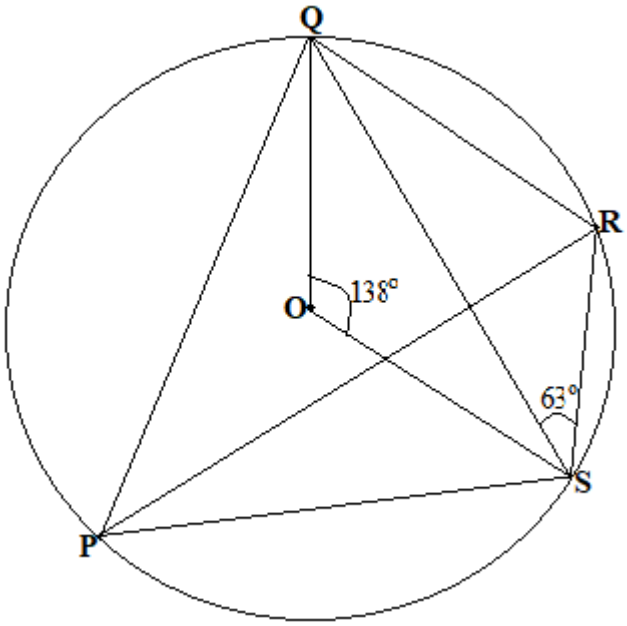


Figure 4

Calculate the size of angle **RQS**. (5 marks)